

Unit 103 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 53.4 ac, BAF 20 variable-radius cruise on a 300 ft grid, 27 plots, 276 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	204 sq ft/ac (SE 3.5 %)	Type	Sierran mixed conifer
Trees per acre	108	CWHR size / density	4 / D (cover 80 %)
QMD	18.6 in	SDI	273 (47 % of max 585)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	196 sq ft/ac 20.6 in	BA QMD	8 sq ft/ac 8.0 in
TPA	85	TPA	23
Volume	40.4 MBF/ac (7341 cu ft)	Volume	3.9 green tons/ac (131 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~153 sq ft/ac BA	Sampling error	7.2 % at 95 % (t = 2.06, n = 27)
Species (BA)	WF 65, PP 58, DF 37, IC 27, SP 18	Stratum standard	40 % - MEETS; 27 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	4	10.6	3.0	45	0.2
8-11.9	19	26.0	14.1	258	1.4
12-15.9	36	24.8	26.7	700	3.8
16-19.9	41	17.7	30.4	878	4.8
20-23.9	46	13.0	34.1	1211	6.7
24-27.9	32	6.5	23.7	898	4.9
28-31.9	26	4.0	19.3	721	4.0
32-35.9	17	2.0	12.6	595	3.3
36-39.9	15	1.4	11.1	539	3.0
40-43.9	12	1.0	8.9	436	2.4
44-47.9	8	0.5	5.9	329	1.8
48-51.9	10	0.5	7.4	412	2.3
52-55.9	2	0.1	1.5	92	0.5
56-59.9	3	0.1	2.2	132	0.7
60-63.9	5	0.2	3.7	226	1.2
All	276	108.4	204.4	7472	41.1

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	88	33.8	65.2	18.8	2533	13.9
PP	78	33.6	57.8	17.8	2018	11.1
DF	50	18.3	37.0	19.2	1363	7.5
IC	36	13.6	26.7	19.0	956	5.3
SP	24	9.1	17.8	18.9	601	3.3
All	276	108.4	204.4	18.6	7472	41.1

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 103: plots on the 300 ft grid (orange = QA flag)

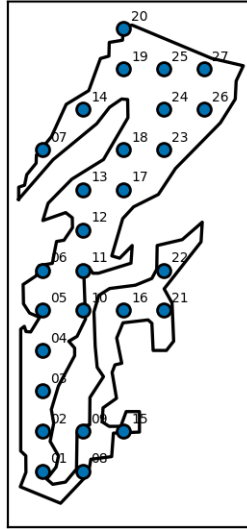


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
103-01	9	180	9	ok
103-02	11	220	174	ok
103-03	14	280	103	ok
103-04	10	200	157	ok
103-05	12	240	109	ok
103-06	8	160	132	ok
103-07	12	240	138	ok
103-08	10	200	127	ok
103-09	7	140	103	ok
103-10	9	180	159	ok
103-11	11	220	141	ok
103-12	8	160	95	ok
103-13	10	200	81	ok
103-14	13	260	113	ok
103-15	11	220	101	ok
103-16	9	180	70	ok
103-17	10	200	55	ok
103-18	12	240	56	ok
103-19	9	180	76	ok
103-20	7	140	62	ok
103-21	7	140	26	ok
103-22	10	200	127	ok
103-23	11	220	70	ok
103-24	12	240	63	ok
103-25	12	240	67	ok
103-26	12	240	68	ok
103-27	10	200	34	ok